

Development of a Portable 3D Pulling Illusion System Using the 3D Vibration Motor “XReactor”

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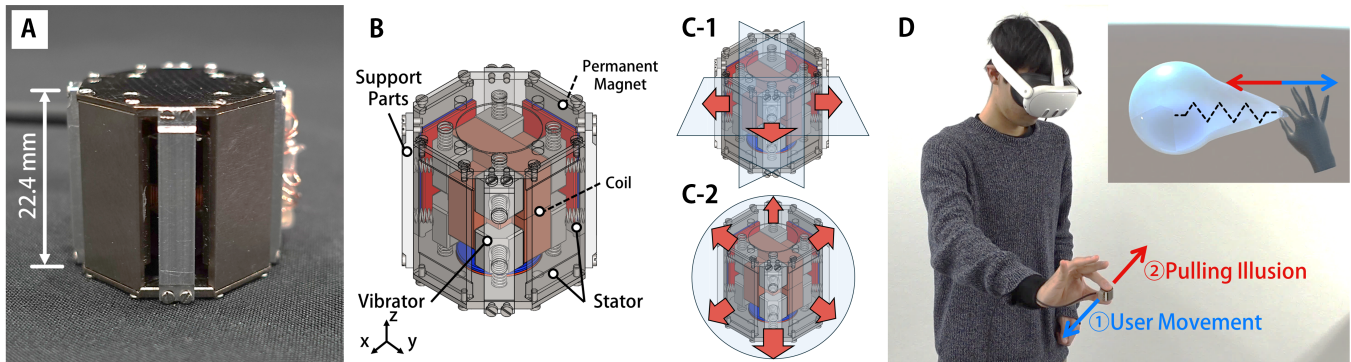


Figure 1: (A)Overview and (B)Components of XReactor. It drives not only 1D but also (C-1)2D and (C-2)3D. (D)It provides 3D haptics in response to user movements.

Abstract

The realization of 3D haptics will expand tactile expression and advance extended reality (XR) applications. This paper describes “XReactor,” a motor that enables 3D vibration in any direction with a single vibrator, and the 3D haptics application for virtual reality (VR) using it. Inside the XReactor, the vibrator consists of three orthogonal coils, whereby the generated electro-magnetic force achieves 3D actuation. By eliminating the complex coordinated control of multiple vibrators, which are one of the major challenges in conventional 3D haptics, this approach allows for a compact and simple implementation. Furthermore, by combining it with VR, it provides 3D tactile feedback when users interact with virtual objects.

CCS Concepts

• Human-centered computing → Haptic devices.

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ACM ISBN 979-8-4007-2549-4/26/07
<https://doi.org/10.1145/3799826.3812422>

Keywords

3D Haptics, Vibration, 3D vibration motor, Virtual reality

ACM Reference Format:

Soya Kokubo, Akira Heya, Riko Ogawara, Shinji Hatano, Masaaki Ando, and Tsuyoshi Inoue. 2026. Development of a Portable 3D Pulling Illusion System Using the 3D Vibration Motor “XReactor”. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3799826.3812422>

1 Introduction

In XR technology, representing reality with sophistication requires not only visual and auditory information but also haptic feedback. Specifically, realizing 3D haptics promises to expand tactile expression ranging from the shape to the texture of 3D virtual objects and enhances immersion through more realistic tactile feedback. In 3D haptics, the use of vibrotactile haptics enables device miniaturization and portability. For example, devices such as WAVES [Culbertson et al. 2017] and HaptiX [Phascher et al. 2023] use multiple vibration motors. However, having multiple vibrator elements and vibration centers involves difficult and precise synchronization, which limits the realization of 3D haptics.

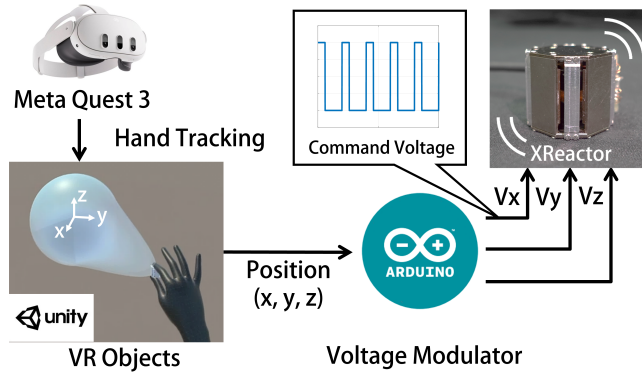


Figure 2: 3D Haptics Systems from VR to XReactor

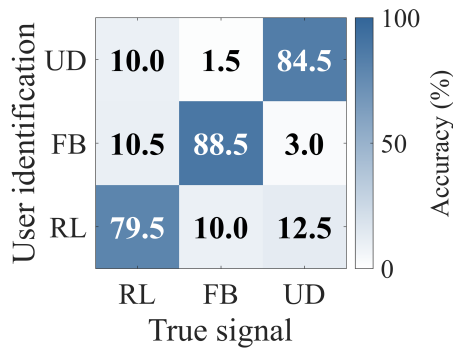


Figure 3: Results of 3Dof Pilling Illusion Test (LR: Right-Left, FB: Forward-Backward, UD: Up-Down)

To address this challenge, we propose “XReactor,” which is based on a 3D vibration motor [Heya et al. 2023], with a single vibrator and a unified vibration center. This eliminates the need for complex synchronization control. Moreover, by integrating VR, we built a system enabling 3D haptic feedback for virtual objects. We demonstrate a 3-DOF pulling illusion, a circular orbit traction via a 2D pulling illusion, and 3D haptic representation with VR.

2 Prototype

Overview. As shown in Figure 1 (B), XReactor primarily consists of an outer stator and an inner vibrator. It measures 22.4 mm × 22.4 mm × 22.4 mm and weighs 37.6 g. Coil springs suspend the vibrator, aligning the actuation and stator centers.

Driving. Principle. XReactor generates 3D vibrations via electromagnetic forces between its internal coils and permanent magnets. When current flows through the coil, the vibrator interacts with the permanent magnets to generate driving forces. As shown in Figure 1 (B), the three coils are arranged orthogonally to eliminate crosstalk. Therefore, by modulating the currents through the three coils, XReactor can be driven in any 3D direction.

Application. Integrating VR with XReactor, as shown Figure 2, enables a more realistic tactile experience than conventional methods. The system uses Meta Quest 3’s hand tracking to map the

physical XReactor to its virtual model in Unity. Based on virtual interactions, Unity sends signals via Arduino to apply command voltages to the coils. This provides 3D haptic feedback synchronized with both user actions and virtual object movements.

3 Demonstration

In SIGGRAPH 2026 Emerging Technology, we demonstrate the following experience using XReactor, applying the pulling illusion through asymmetric vibration [Amemiya and Maeda 2008].

The 3-DOF Pulling Illusion. Users grasp the XReactor unit between the thumb and index finger. By applying an asymmetric vibration voltage based on rectangular waves, the device induces a pulling illusion along the x, y, and z axes. By selectively driving the coils, users can experience the pulling illusion in one to three dimensions. As shown in Figure 3, our verification demonstrated that the pulling illusion is achievable in any direction.

Circular Orbit Traction. XReactor generates a circular pulling illusion by rotating the asymmetric vibration axis. Modulating two of the three coils with composite waveforms achieves this effect. Furthermore, by varying the combination of active coils, circular orbit traction can be rendered in the x-y, y-z, and x-z planes.

3D Pulling with VR. Users grasp XReactor and interact with a virtual balloon. By pushing or pulling the balloon membrane in VR, users experience a virtual elastic force. This is achieved by generating a pulling illusion opposite to the stretch direction. By mapping the degree of stretch to the input, the system provides a stronger perceived rebound force as the deformation.

References

- Tomohiro Amemiya and Taro Maeda. 2008. Asymmetric Oscillation Distorts the Perceived Heaviness of Handheld Objects. *IEEE Transactions on Haptics* 1, 1 (January–June 2008), 1312–1320. doi:10.1109/TOH.2008.13
- Heather Culbertson, Julie M. Walker, Michael Raitor, and Allison M. Okamura. 2017. WAVES: A Wearable Asymmetric Vibration Excitation System for Presenting Three-Dimensional Translation and Rotation Cues. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (May 6–11, 2017)* (Denver, Colorado) (CHI '17). ACM Inc., New York, NY, USA, 4972–4982. doi:10.1145/3025453.3025741
- Akira Heya, Ryosuke Nakamura, and Katsuhiro Hirata. 2023. Development of Compact 3-Degree-of-Freedom Oscillatory Actuator. *Journal of Robotics and Mechatronics* 35, 5 (Oct. 2023), 1312–1320. doi:10.20965/jrm.2023.p1312
- Max Phascher, Til Franzen, Kirill Kronhardt, Uwe Gruenefeld, Stefan Schneegass, and Jens Gärken. 2023. HaptiX: Vibrotactile Haptic Feedback for Communication of 3D Directional Cues. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (April 23–28, 2023)* (Hamburg, Germany) (CHI EA '23). ACM, New York, NY, USA. doi:10.1145/3544549.3585601